

20260924

- Datasets: array_5388, meta_5688
- Chromosomes: 2, 3, 5, 16

Objective

- Infer chromosome-specific ancestral recombination graphs (ARGs) from phased genotype data using tsinfer, followed by estimation of node ages using tsdate with the `variational_gamma` dating method to produce dated `.trees`.
- The analysis is performed independently for chromosomes 2, 3, 5, and 16 rather than constructing a whole-genome tree sequence.

Overall Workflow

1. Input Data:
 - i. array_5388 (MEGA + GDA imputed data)
 - ii. meta_5688 (TOPMed-imputed data)
2. `.vcf.gz` preprocessessing + ancestral-state annotation
 - i. ancestral state file:
/project2/chia657_28/resources/ancestral_alignments/human_ancestor_GRCh38_e114 , copied to /project2/mahj_1946/variational_gamma/Data/homo_sapiens_ancestor_{2, 3, 5, 16}.fa
3. `.vcz` intermediate output
 - i. Processess involves **chunking chromosome into 10Mb increments for parallel processing + memory load reduction
4. `bio2zarr` + Zarr variant data structure
5. `.vcz` validation and **duplicate-position** filtering
6. `tsinfer`
7. Chromosome-specific `.trees`
8. `tsdate preprocess_ts()`
9. `tsdate date(method="variational_gamma")`
10. Final output: dated chromosome-specific `.trees`

The `.vcz` directories are retained as the primary intermediate genotype representation. Existing `.vcz` directories are used directly for the current `tsinfer` analysis rather than being regenerated from the

original .vcfs .

De-duplication step

- The .vcz directories already contain the variant_AA ancestral-state field corresponding to the VCF AA annotation, so ancestral states do not need to be regenerated from the ancestral FASTA/BED resources for this analysis.

VCZ filtering

- The existing array_5388 and meta_5688 .vcz directories contained duplicate genomic positions. Because tsinfer requires variant_position to be strictly increasing, duplicate records **must be** removed before inference.
 - Note that sorting the genomic positions does not inherently remove duplicate records, but rather puts them in adjacent indices in the position array.
- The filtering step creates: [meta-imputed_MEC_TOPMed_ / array_5388]chr{2, 3, 5, 16}.dedup.vcz
- The filtering script retains the **first variant record** at each genomic position and removes subsequent records at that position.

For example:

```
20665 C>A
20665 C>T
```

becomes:

```
20665 C>A
```

- This is a pragmatic preprocessing step intended to satisfy the strict positional ordering required by tsinfer.

Python scripts

vcz_to_trees.py

- This is the primary inference and dating script.
- The script:
 - Opens an existing .vcz dataset.
 - Constructs a tsinfer.formats.VariantData object.

- Uses the existing ancestral-state field (`variant_AA`).
- Runs `tsinfer.infer()` .
- Preprocesses the inferred tree sequence using `tsdate.preprocess_ts()` .
- Dates the tree sequence using:

```
tsdate.date(
    dating_ts,
    mutation_rate=1.5e-8,
    method="variational_gamma",
)
```

- Writes the resulting dated tree sequence to a `.trees` file.
- The relevant conceptual transition is therefore:

1. `.vcz`
2. `VariantData`
3. `tsinfer`
4. `inferred .trees`
5. `tsdate.preprocess_ts()`
6. `tsdate variational_gamma`
7. `dated .trees`

- The default mutation rate being used is: 1.5×10^{-8} mutations/site/generation

`filter_vcz_duplicates.py`

- This script is used to repair existing `.vcz` files containing duplicate variant positions.
- The script:
 - Reads `variant_position` .
 - Identifies consecutive records with identical positions.
 - Retains the **first record** at each position.
 - Removes subsequent duplicate records.
 - Copies the remaining variant-indexed arrays—including genotype and allele information—to a new `.vcz` .
 - Preserves non-variant arrays and `zarr` metadata.
 - Verifies that the resulting `variant_position` array is strictly increasing.
- This produces a new `.dedup.vcz` directory without modifying the original `.vcz` .
- The script processes large arrays in chunks to avoid loading the full genotype matrix into memory.

Current analysis state

- The current workflow status for each dataset is:
- array_5388
 - chr2 — [**.vcf.gz**] — ancestral + sorted vcf.gz — .vcz — dedup.vcz — dated .trees
 - Note: There is an unresolved issue for chr2 sorting + implementing the ancestral state array concerning memory. I'm using the oneweek partition to test.
 - chr3 — .vcf.gz — [**ancestral + sorted vcf.gz**] — .vcz — dedup.vcz — dated .trees
 - chr5 — .vcf.gz — ancestral + sorted vcf.gz — [**.vcz**] — dedup.vcz — dated .trees
 - chr16 — .vcf.gz — ancestral + sorted vcf.gz — .vcz — [**dedup.vcz**] — dated .trees
- meta_5388
 - chr2 — .vcf.gz — [**ancestral + sorted**] — .vcz — dedup.vcz — dated .trees
 - chr3 — .vcf.gz — ancestral + sorted vcf.gz — [**.vcz**] — dedup.vcz — dated .trees
 - chr5 — .vcf.gz — ancestral + sorted vcf.gz — .vcz — [**dedup.vcz**] — dated .trees
 - chr16 — .vcf.gz — ancestral + sorted vcf.gz — .vcz — [**dedup.vcz**] — dated .trees